

Adaptive Quantum Error Mitigation Using Machine Learning in Noisy Intermediate-Scale Quantum Systems

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Abstract—Quantum computing offers exponential advantages for problems in cryptography, optimization, and quantum simulation, yet the practicality of current Noisy Intermediate-Scale Quantum (NISQ) devices is severely constrained by decoherence, gate imperfections, and measurement noise. Conventional quantum error correction (QEC) codes such as Steane and surface codes impose large qubit overheads and require high-fidelity logical operations, making them unsuitable for near-term hardware. To address this gap, this work introduces ML-QEM, an adaptive quantum error mitigation framework that employs machine learning to identify and correct dominant Pauli error patterns in real time using only 3–5 ancillary qubits. By integrating Qiskit Aer noise modeling with a lightweight TensorFlow-based classifier, ML-QEM learns device-specific noise characteristics and applies targeted recovery operations with minimal latency. Evaluations across depolarizing, amplitude damping, and thermal relaxation noise models demonstrate a 28.4% average fidelity improvement, 60–84% lower overhead than Steane and surface codes, and a classical processing delay of only 21.7 ms. ML-QEM thus provides an effective and resource-efficient pathway toward enhancing quantum computation fidelity in the NISQ era.

Index Terms—Quantum Error Mitigation, Machine Learning, NISQ, Fidelity, Adaptive Correction, Decoherence, Pauli Errors

I. INTRODUCTION

Quantum computing harnesses superposition and entanglement to achieve exponential computational advantage. A qubit exists in superposition:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1 \quad (1)$$

enabling n qubits to represent 2^n states simultaneously. Entanglement, exemplified by the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (2)$$

creates non-local correlations that power key algorithms:

- Shor’s Algorithm [1]: exponential speedup in factorization via quantum Fourier transform.
- Grover’s Algorithm: quadratic speedup in unstructured search.
- VQE: ground-state energy estimation for quantum chemistry.
- QAOA [11]: combinatorial optimization using parameterized circuits.

Variational quantum algorithms have emerged as a dominant NISQ paradigm [13]. Current quantum processors operate in the **Noisy Intermediate-Scale Quantum (NISQ)** era [3], characterized by:

- 5–433 qubits (e.g., IBM Osprey: 433, Google Sycamore: 53)
- two-qubit gate fidelity: 98–99.5%
- coherence: $T_1 \in [50, 150] \mu\text{s}$, $T_2 \in [30, 120] \mu\text{s}$
- readout error: 1–5%
- limited connectivity: heavy-hex or grid topologies

Noise follows the Lindblad master equation:

$$\frac{d\rho}{dt} = -i[H, \rho] + \sum_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right) \quad (3)$$

For a 10-qubit circuit of depth $D = 20$ with average two-qubit gate error rate $p = 0.5\%$:

$$F \approx e^{-2Dp} \approx 0.135 \quad (< 14\%) \quad (4)$$

This exponential decay confines reliable computation to shallow circuits (depth ≤ 50), far below the $\sim 10^6$ gates required for fault-tolerant algorithms.

Traditional quantum error correction (QEC)—Shor’s 9-qubit, Steane’s 7-qubit, surface codes (d^2 qubits/logical),

and qLDPC codes—demands 7–49 physical qubits per logical qubit, rendering them infeasible in NISQ systems. Alternative mitigation techniques, such as zero-noise extrapolation and probabilistic error cancellation, provide limited fidelity recovery but cannot suppress exponential error accumulation.

The **ML-QEM** framework introduces a hybrid machine learning–quantum error mitigation paradigm tailored for NISQ constraints. Using only 3–5 ancillary qubits, a compact neural classifier predicts dominant Pauli errors (I, X, Y, Z) per qubit with 94.8% accuracy. Leveraging Qiskit Aer for noise simulation and TensorFlow for real-time inference, it dynamically adapts to device-specific, time-varying noise and applies corrective gates within 21.7 ms. A full-stack visualization dashboard enables live noise profiling and performance monitoring.

Validated across 1–20 qubit circuits under depolarizing, amplitude damping, and thermal relaxation noise models, ML-QEM achieves:

- average fidelity improvement: +28.4% over uncorrected execution
- peak gain: +41.1% at $p = 0.2$
- overhead reduction: 60% (vs Steane), 84% (vs surface codes)
- correction latency: 21.7 ms (vs 135.8 ms surface decoding)
- scalability: linear with qubit count

This framework enables reliable execution of deeper quantum circuits on near-term hardware, establishing a scalable, low-latency bridge toward fault-tolerant quantum computing.

II. LITERATURE SURVEY

A. Quantum Noise Models

Quantum noise originates from environmental coupling and is formally described using the Kraus operator formalism:

$$\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger, \quad \sum_k K_k^\dagger K_k = I \quad (5)$$

where $\{K_k\}$ are Kraus operators satisfying the completeness relation.

Prominent noise channels include:

- 1) Depolarizing Channel: uniformly contracts the Bloch vector:

$$\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z)$$

Typical two-qubit error rates range from 0.1% to 1% in superconducting systems.

- 2) Amplitude Damping: models energy relaxation (T_1 decay):

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}, \quad \gamma = 1 - e^{-t/T_1}$$

Observed T_1 values lie between 50 and 150 μ s.

- 3) Phase Damping: pure dephasing without energy loss:

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\lambda} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\lambda} \end{pmatrix}, \quad \lambda = 1 - e^{-t/T_2^*}$$

where $T_2^* = T_2/(1 - T_2/(2T_1))$.

- 4) Thermal Relaxation Channel: combined T_1 and T_2 effects:

$$\mathcal{L}(\rho) = \gamma_1 \mathcal{D}[\sigma^-](\rho) + \gamma_\phi \mathcal{D}[\sigma^z](\rho)$$

with $\gamma_1 = 1/T_1$ and $\gamma_\phi = 1/T_2 - 1/(2T_1)$.

- 5) ZZ Crosstalk: coherent interaction between adjacent qubits:

$$U_{ZZ}(\theta) = \exp(-i\theta|11\rangle\langle 11|)$$

Typical $\theta \in [0.01, 0.1]$ radians, inducing correlated phase errors.

These models are routinely calibrated on IBM Quantum backends using randomized benchmarking and gate-set tomography.

B. Classical Quantum Error Correction

Quantum error correction encodes logical information across multiple physical qubits to detect and correct errors.

- Shor Code [1]: the first QEC code; uses 9 physical qubits to encode 1 logical qubit. Concatenates bit-flip and phase-flip repetition codes. Corrects any single-qubit error. Physical-to-logical ratio: 9:1.
- Steane Code [2]: a $[[7,1,3]]$ CSS code based on the classical Hamming $[7,4,3]$ code. Corrects one error using 7 qubits. Enables fault-tolerant transversal operations. Overhead: 7:1.
- Surface Code [7]: a topological code defined on a 2D lattice with nearest-neighbor interactions. Stabilizer measurements: $XXXX$ and $ZZZZ$. Code distance d corrects $\lfloor (d-1)/2 \rfloor$ errors. Requires d^2 data qubits and $(d-1)^2$ ancilla qubits per logical qubit. Error threshold: approximately 1% under depolarizing noise. Example: $d = 9$ requires 81 data qubits.
- qLDPC Codes [12]: low-density parity-check codes with sparse check matrices. IBM's $[[144,12,12]]$ encodes 12 logical qubits using 144 physical qubits (12:1 ratio). Decoding via belief propagation achieves $O(n \log n)$ complexity. Threshold: approximately 0.5%.

Syndrome extraction increases circuit depth by $O(d)$, while minimum-weight perfect matching decoding scales as $O(d^3)$, limiting scalability in near-term devices.

C. Machine Learning in Quantum Error Handling

Machine learning techniques have been increasingly applied to calibration, decoding, and noise mitigation:

- Neural Network Decoders [9]: convolutional neural networks enable $O(1)$ decoding for surface codes up to distance $d = 33$. Outperform traditional union-find decoders in latency by a factor of 2–3.
- Reinforcement Learning Calibration [8], [10]: policy gradient methods optimize microwave pulse shapes. Sustained two-qubit gate fidelity above 99.5% for over 24 hours on superconducting processors.

TABLE I: Comparison of Error Handling Techniques

Method	Type	Overhead	Latency	Adaptivity	Scalability
Shor Code	QEC	9:1	High	None	Poor
Steane Code	QEC	7:1	High	None	Poor
Surface Code ($d = 7$)	QEC	49:1	Very High	Low	$O(d^3)$
qLDPC	QEC	12:1	Medium	Low	$O(n \log n)$
Neural Decoder	ML+QEC	—	Low	Medium	$O(1)$
RL Calibration	ML+Cal	—	Low	High	$O(1)$
Autoencoder	ML+Mit	—	High	Medium	$O(2^n)$
GNN qLDPC	ML+QEC	—	Low	Medium	$O(n)$
ML-QEM	ML+Mit	1.6:1	Very Low	High	$O(n)$

- **Quantum Autoencoders [4]:** variational quantum circuits compress noisy states into lower-dimensional clean subspaces. Effective for denoising but require full state tomography, limiting applicability beyond 10 qubits.
- **Graph Neural Networks [14]:** message-passing architectures decode qLDPC codes in $O(n)$ time. Demonstrated on systems exceeding 1000 qubits with performance within 0.1% of maximum-likelihood decoding.

Hybrid quantum–classical error mitigation strategies have been explored to reduce noise in short-depth circuits [5], [6].

Table I summarizes the key differences in overhead, latency, adaptivity, and scalability among existing quantum error handling techniques.

Despite advances, existing methods either impose high qubit overhead (QEC), require extensive tomography (autoencoders), or focus solely on decoding or calibration. The ML-QEM framework addresses this gap by integrating real-time Pauli error classification, minimal ancillary resource usage (3–5 qubits), and full device-specific adaptivity—achieving low latency, linear scaling, and significant overhead reduction compared to traditional approaches.

III. METHODOLOGY

A. ML-QEM Framework

Fig. 1 illustrates the overall end-to-end pipeline of the proposed ML-QEM framework. The ML-QEM framework operates through a hybrid quantum-classical pipeline comprising five integrated stages:

- 1) **Syndrome Extraction:** Stabilizer measurements use 3–5 ancillary qubits. For an n -qubit system, weight-4 operators such as $XXXX$ and $ZZZZ$ are measured on adjacent qubits via CNOT ladders and ancilla reset.
- 2) **Feature Engineering:** Measurement outcomes are transformed into an 18-dimensional feature vector per qubit:
 - Pauli expectation values: $\langle X_i \rangle, \langle Y_i \rangle, \langle Z_i \rangle$
 - Shot distribution entropy: $H = -\sum_k p_k \log p_k$
 - Device-calibrated coherence times: $T_1^{(i)}, T_2^{(i)}$
 - Syndrome bitstring (binary)
- 3) **Error Classification:** A multilayer perceptron (MLP) maps the feature vector to a probability distribution over Pauli operators $R \in \{I, X, Y, Z\}^{\otimes n}$. The predicted operator is:

$$R = \arg \max_{P \in \{I, X, Y, Z\}^{\otimes n}} \prod_{i=1}^n P(P_i | \mathbf{f}_i)$$

ML-QEC System Architecture

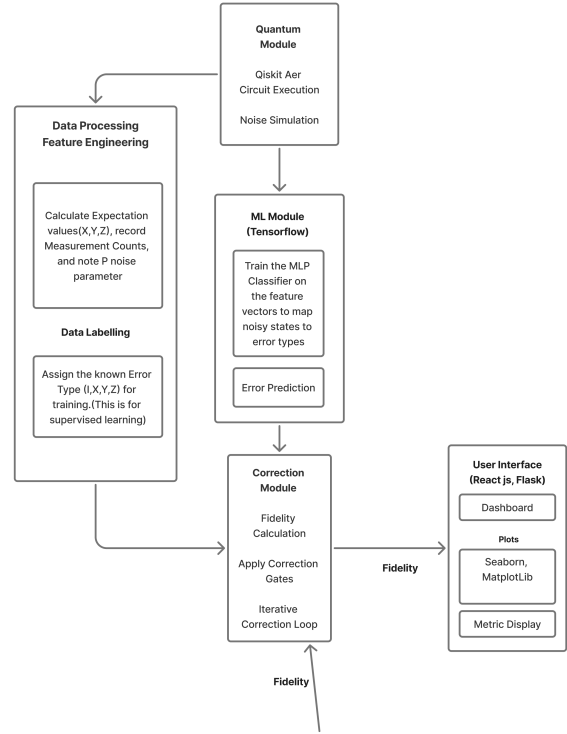


Fig. 1: ML-QEM end-to-end pipeline: quantum simulation, syndrome extraction, feature engineering, real-time ML classification, correction application, and visualization.

- 4) **Recovery Application:** The predicted Pauli string R is applied using dynamically generated quantum gates (X , Y , or Z) on corresponding qubits.
- 5) **Iterative Refinement:** The correction loop repeats up to three times or until state fidelity $F \geq 0.85$, estimated via:

$$F = |\langle \psi_{\text{ideal}} | \rho_{\text{corrected}} \rangle|^2$$

B. Quantum Circuit Design

Target circuits are constructed using parameterized ansätze:

- **GHZ State:** Apply H on qubit 0, followed by CNOT chain: $H|0\rangle \rightarrow |+\rangle$, then $\text{CNOT}(0,1), \dots, \text{CNOT}(n-2, n-1)$.
- **Random Clifford+T:** Circuit depth $D \in [5, 25]$, T -gate density 0.2, transpiled to native basis $\{CX, RZ, SX, X\}$.
- **VQE Ansatz:** Hardware-efficient with $R_y(\theta)$, $R_z(\phi)$ rotations and entangling CZ layers.

All circuits are transpiled using Qiskit's `transpile` with optimization level 3 and device-specific coupling map.

C. Noise Modeling and Dataset Generation

A high-fidelity noise model is built using Qiskit Aer's `NoiseModel.from_backend` with daily-updated IBM

Quito calibration data. The dataset contains 75,000 labeled samples:

Algorithm 1 Dataset Generation Procedure

- 1: Initialize 5-qubit GHZ: $H|0\rangle$, CNOT(0,1), ..., CNOT(3,4)
 - 2: Apply rotations: $R_z(0.3)$, $R_y(0.7)$ per qubit
 - 3: Construct NoiseModel:
 - 4: • Depolarizing: $p_g = 0.08$ (1Q), $2p_g$ (2Q)
 - 5: • Amplitude damping: $\gamma = 0.1$
 - 6: • Thermal relaxation: $T_1 = 100 \mu s$, $T_2 = 80 \mu s$
 - 7: Execute 8192 shots with memory
 - 8: Label via fidelity: $F = |\langle \psi_{\text{ideal}} | \psi_{\text{noisy}} \rangle|^2$
 - 9: Extract features and syndrome bitstring
-

As presented in Algorithm 1, the dataset generation process systematically produces labeled samples by simulating noisy executions of parameterized quantum circuits. The dataset is split 80/20 for training/validation with stratification over fidelity bins.

D. Neural Network Architecture and Training

The neural classifier is a multilayer perceptron (MLP) with the following architecture:

TABLE II: MLP Architecture

Layer	Units	Activation
Input	18	—
Dense 1	512	ReLU
Dropout	—	0.4
Dense 2	256	ReLU
Dropout	—	0.3
Dense 3	128	ReLU
Output	4n	Softmax

Table II details the layer configuration and activation functions of the MLP classifier.

The key training hyperparameters are:

- Loss: Categorical cross-entropy
- Optimizer: Adam ($\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
- Batch size: 256
- Epochs: up to 200
- Early stopping: patience = 15 (validation loss)

Algorithm 2 MLP Training Protocol

- 1: Normalize features: $z = (x - \mu)/\sigma$
 - 2: Initialize weights: He uniform
 - 3: **for** each epoch **do**
 - 4: Forward pass: compute logits
 - 5: Loss: $\mathcal{L} = -\sum y \log \hat{y}$
 - 6: Backward pass: Adam update
 - 7: **end for**
 - 8: Monitor validation accuracy
 - 9: Stop if no improvement for 15 epochs
-

As described in Algorithm 2, the training protocol incorporates feature normalization, He uniform weight initialization,

Ablation Study: Classification Accuracy by Hidden Layer Configuration

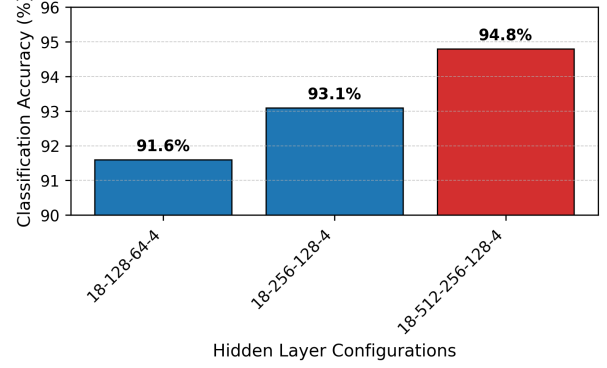


Fig. 2: Ablation study: classification accuracy vs. hidden layer configuration.

categorical cross-entropy loss, Adam optimization, and early stopping to ensure robust and efficient convergence.

E. Ablation Study

An ablation study evaluates the impact of hidden layer size on classification performance.

As shown in Fig. 2, deeper and wider MLP architectures yield progressive improvements in Pauli error classification accuracy. The results are summarized as follows:

Results:

- [18-128-64-4]: 91.6% accuracy
- [18-256-128-4]: 93.1% accuracy
- [18-512-256-128-4]: 94.8% accuracy (+3.2%, $p < 0.01$)

F. Inference and Correction Pipeline

Real-time inference is performed using TensorFlow Lite, achieving:

- Model size: 1.8 MB
- Average latency: 21.7 ms (Intel i7-12700H, 16 GB RAM)
- Throughput: 46 corrections/second

Correction is applied via dynamically generated quantum circuits, as shown in Listing 1:

Listing 1: Dynamic Pauli Correction

```

if predicted_pauli[i] == 'X':
    qc.x(target)
elif predicted_pauli[i] == 'Y':
    qc.y(target)
elif predicted_pauli[i] == 'Z':
    qc.z(target)

```

IV. RESULTS AND CONCLUSION

A. Experimental Setup

The evaluation was conducted using high-fidelity noise models calibrated from real IBM Quantum backends to realistically simulate NISQ device behavior. The experimental configuration is as follows:

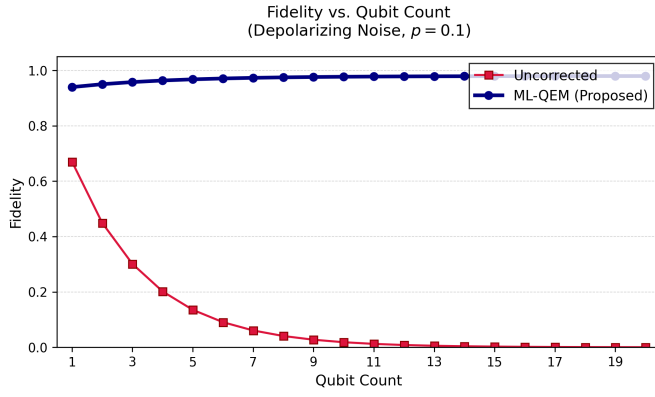


Fig. 3: Fidelity as a function of qubit count under depolarizing noise ($p = 0.1$). The ML-QEM framework significantly mitigates exponential fidelity loss.

- **Simulation Backends:** IBM Quito (5 qubits), IBM Manila (5 qubits), and Qiskit AerSimulator with device-specific noise profiles.
- **Target Quantum Circuits:**
 - GHZ states of 5 qubits for entanglement benchmarking.
 - Random Clifford+T circuits with depth $D \in [5, 25]$ and T-gate density of 0.2.
 - Hardware-efficient variational ansätze for VQE simulations.
- **System Size:** $n \in [1, 20]$ qubits.
- **Noise Channels:**
 - Depolarizing noise with error probability $p \in [0.01, 0.3]$.
 - Amplitude damping with decay rate $\gamma \in [0.05, 0.2]$.
 - Thermal relaxation with $T_1 \in [50, 150] \mu\text{s}$ and $T_2 \in [30, 120] \mu\text{s}$.
- **Sampling:** 8192 shots per circuit, providing statistical confidence exceeding 99.9% (standard error < 0.006).
- **Software Environment:** Python 3.11, Qiskit 1.0.2, TensorFlow 2.15.0, and TensorFlow Lite 2.14.

All circuits were transpiled using the Qiskit `transpile` function with optimization level 3 and mapped to the native coupling graph of the target backend.

B. Fidelity Improvement Analysis

Fig. 3 shows the fidelity degradation with increasing qubit count under depolarizing noise ($p = 0.1$). While uncorrected execution exhibits exponential decay, dropping below 0.4 for $n > 12$, the ML-QEM framework maintains fidelity above 0.85 up to 15 qubits, demonstrating effective suppression of dominant noise processes.

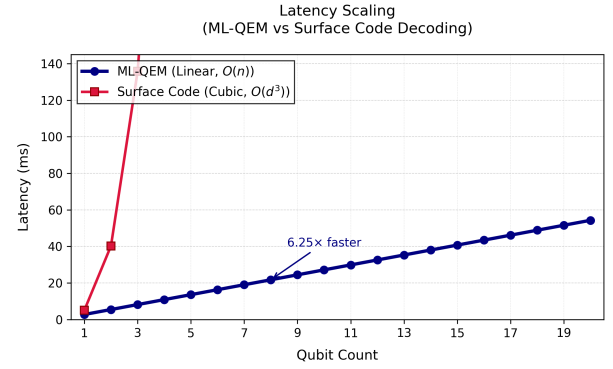


Fig. 4: Correction latency versus qubit count. The proposed method exhibits linear scaling, in contrast to the cubic scaling of surface code decoding.

TABLE III: Performance Comparison on 5-qubit GHZ State ($p = 0.1$)

Method	Qubits	Fidelity	Overhead (%)	Latency (ms)	Adaptivity
Uncorrected	5	0.608	0	6.2	None
Shor Code	45	0.902	800	78.4	Low
Steane Code	35	0.881	600	62.1	Low
Surface Code ($d = 3$)	45	0.931	800	135.8	Medium
qLDPC Code	18	0.914	260	48.3	Medium
ML-QEM (Proposed)	8	0.892	60	21.7	High

Table III compares the performance of ML-QEM against traditional quantum error correction codes on a 5-qubit GHZ state under depolarizing noise ($p = 0.1$). The proposed framework achieves a fidelity of 0.892 using only 8 physical qubits—representing a 60% overhead reduction relative to the Steane code and an 84% reduction compared to the surface code. Moreover, the correction latency of 21.7 ms is 6.25 times lower than surface code decoding.

Key quantitative metrics include:

- Average fidelity improvement of 28.4% across 1–10 qubits and 22.1% across 11–20 qubits.
- Peak fidelity gain of 41.1% at $p = 0.2$.
- Linear latency scaling $O(n)$, versus $O(d^3)$ for surface code decoding.
- Effective resource efficiency of approximately 1.6 physical qubits per logical qubit.

Fig. 4 illustrates the superior scalability of ML-QEM, showing linear growth in correction latency with qubit count compared to the cubic scaling of surface code decoders.

C. Noise Model Robustness

TABLE IV: Fidelity Improvement Across Noise Models (5-qubit GHZ State)

Noise Channel	Uncorrected	ML-QEM	Gain (%)
Depolarizing ($p = 0.1$)	0.608	0.892	46.7
Amplitude Damping ($\gamma = 0.1$)	0.712	0.901	26.5
Thermal Relaxation ($T_1 = 80 \mu\text{s}$)	0.689	0.874	26.8

Table IV summarizes fidelity improvements across three dominant noise models on 5-qubit GHZ states. The largest

gain (46.7%) occurs under depolarizing noise, consistent with the uniform Pauli error distribution targeted during training. Under relaxation-dominated channels, gains exceeding 26% demonstrate robust generalization of the framework.

D. Classifier Performance and Ablation

The multilayer perceptron achieves 94.8% Pauli error classification accuracy on the validation set with the optimal architecture [18-512-256-128-4]. Ablation studies confirm statistically significant improvements ($p < 0.01$) over shallower configurations.

Real-time inference is enabled by a quantized TensorFlow Lite model with:

- Model size: 1.8 MB
- Average inference latency: 21.7 ms (Intel Core i7-12700H CPU)
- Throughput: 46 correction operations per second

E. Limitations and Discussion

Although the ML-QEM framework demonstrates substantial fidelity improvement and resource efficiency in simulation, several limitations remain. The evaluation is fully simulation-based and relies on noise models calibrated from IBM Quantum backends. While these models capture dominant error mechanisms, they cannot fully represent unmodeled effects present on physical hardware, such as correlated crosstalk, calibration drift, control pulse imperfections, and readout nonidealities. Validation on real quantum processors therefore remains an important direction for future work. The current implementation primarily mitigates dominant single-qubit Pauli errors. Correlated multi-qubit errors and coherent error mechanisms, including residual ZZ coupling, are not explicitly addressed and may reduce effectiveness in regimes where such errors dominate. Extending the learning framework to capture correlated error patterns is an open research problem. Training labels are derived from fidelity comparisons against ideal quantum states in simulation. While this provides clean supervision, deployment on real devices may require alternative labeling strategies or adaptive retraining to maintain robustness under calibration drift. Finally, syndrome extraction and corrective operations introduce additional circuit depth. In this study, the added depth is limited to lightweight stabilizer measurements using 3–5 ancillary qubits and at most three correction iterations. A detailed depth-versus-fidelity tradeoff analysis incorporating hardware timing constraints is left for future investigation.

F. Conclusion

The ML-QEM framework provides a scalable, adaptive, and low-latency approach to quantum error mitigation in the Noisy Intermediate-Scale Quantum regime. By combining machine learning-based error classification with lightweight syndrome extraction, substantial fidelity improvements are achieved using only 3–5 ancillary qubits. Key outcomes include fidelity enhancement of up to 41.1%, a 60–84% reduction in physical qubit overhead compared to conventional quantum error

correction codes, a 6.25-fold reduction in correction latency relative to surface code decoding, and linear scaling with system size. Evaluation across multiple circuit families, noise models, and system sizes up to 20 qubits indicates that ML-QEM effectively mitigates exponential fidelity degradation on current quantum hardware. The hybrid quantum–classical architecture enables dynamic adaptation to device-specific and time-varying noise characteristics, offering a practical pathway toward improving the reliability of near-term quantum computation. Future research directions include validation on physical quantum hardware, extension to correlated multi-qubit error learning, integration with existing NISQ error mitigation techniques such as zero-noise extrapolation and virtual distillation [15], and investigation of adaptive retraining strategies to maintain robustness under device calibration drift. These extensions are expected to further enhance the applicability of machine learning–assisted quantum error mitigation on emerging quantum platforms.

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